CAPSTONE — DATA DICTIONARY

Subscription & Churn — Relational Dataset

Customer Subscription & Churn Analysis

Five related tables — every column, its type, and what it means.

1 fact + 4 dimensions • **Build a star schema** • **Synthetic teaching dataset**

> **NOTE**
> **About this dataset.** A synthetic teaching dataset delivered as **five related CSV files** — one fact table and four dimension tables. You load all five, clean them, and model them into a star schema (plus a Date table you build yourself). The tables below describe each column's **intended** clean meaning; the 'watch for' section flags the quality issues you must fix.

The five tables

Table	Role	What it holds
`Subscriptions`	Fact	One row per customer — the subscription facts and the churn outcome.
`Customers`	Dimension	Descriptive customer attributes (name, date of birth, gender).
`Plans`	Dimension	The three subscription tiers and their base price.
`Regions`	Dimension	The customer regions and country.
`PaymentMethods`	Dimension	The payment methods and their type.

> **KEY CONCEPT**
> **The shape is a star.** Subscriptions sits in the middle; each dimension joins to it on its key — `CustomerID`, `PlanID`, `RegionID`, `PaymentMethodID`. You also build a Date table and relate it to `SignupDate`.

Subscriptions (fact)

Column	Type	Meaning
`CustomerID`	Text (key)	Links to Customers. One row per customer.
`PlanID`	Whole number (FK)	Links to Plans.
`RegionID`	Whole number (FK)	Links to Regions.
`PaymentMethodID`	Whole number (FK)	Links to PaymentMethods.
`SignupDate`	Date	When the customer subscribed (dd/mm/yyyy); relate to your Date table.

Column	Type	Meaning
ContractType	Categorical	Billing commitment: Monthly or Annual.
MonthlyCharge	Decimal	Amount billed per month.
TenureMonths	Whole number	Months subscribed.
AutoRenew	Categorical	Auto-renew on: Yes or No.
SupportTickets	Whole number	Support tickets raised.
AvgUsageHours	Decimal	Average monthly usage hours.
LastLoginDays	Whole number	Days since last login (higher = less active).
Churned	Categorical (target)	Has the customer cancelled: Yes or No.

Customers (dimension)

Column	Type	Meaning
CustomerID	Text (key)	Primary key; links to Subscriptions.
CustomerName	Text	Customer's full name.
DateOfBirth	Date	Date of birth (dd/mm/yyyy) — derive Age from this.
Gender	Categorical	Male or Female.

Plans / Regions / PaymentMethods (dimensions)

Table	Columns
Plans	PlanID, PlanName (Basic/Standard/Premium), BaseMonthlyPrice
Regions	RegionID, Region (North/South/East/West), Country
PaymentMethods	PaymentMethodID, MethodName, MethodType (Card/Bank/Digital)

> **NOTE**
> **These three are clean reference tables** — as lookup tables usually are. The cleaning work is in the fact and the Customers dimension.

Data-quality issues to watch for

The raw files are not clean. Expect to deal with each of these (they are part of the assessment):

In Subscriptions (fact)

- **Duplicate customers** — some CustomerIDs appear more than once.
- **Missing values** — blanks in AvgUsageHours, LastLoginDays, and some foreign keys (PaymentMethodID, RegionID).
- **Outliers** — impossible values such as a tenure of 999 months or a charge of 9999.
- **Inconsistent categories** — ContractType, AutoRenew and Churned each spelled several ways (e.g. 'Monthly'/'monthly'/'M'; 'Yes'/'Y'; 'No'/'N'/'0').

- **Wrong data type** — some MonthlyCharge values carry a '$' (forcing text); SignupDate may import as text unless you set its type/locale.
- **Orphan foreign keys** — a few rows reference a PlanID or RegionID that does not exist in the dimension; these show blank once relationships are built.

In Customers (dimension)

- **Duplicate rows** — a few CustomerIDs appear twice; a dimension key must be unique or the relationship is ambiguous.
- **Missing DateOfBirth** and inconsistent Gender (Male/M, Female/F/female).
- **Trailing spaces** in some CustomerName values.

COMMON PITFALL

Clean and model before you analyse. Inconsistent categories split a group into several; a text MonthlyCharge can't be summed; a duplicate dimension key breaks the relationship; and orphan keys silently drop rows from a dimension's totals. Fixing these first is what makes your numbers correct.